

Food and calorie intake tracking using phone (Automated Food Calorie Estimation Using Image-Based Volume Analysis)

Group No: 20 — Project ID: Ind-21

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Problem Description

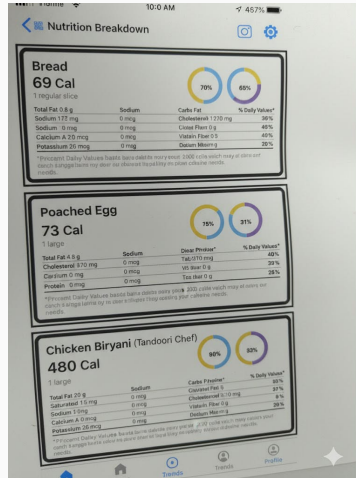
- Accurate calorie estimation is critical for dietary monitoring and health management
- Manual calorie logging is time-consuming and error-prone



Figure 1: Problem faced by user

Problem Description

- Solution: **Image-based approaches offer a scalable alternative**



Problem Description: Requirements

- Take an image of the food
- Identify the name of the food
- Estimate calorie information
- Extract other nutritional components:
 - Carbohydrates
 - Fat
 - Protein etc

Problem Description : Motivation

Why does this research matter to the world?

- While identifying food is easy for AI, **estimating its volume** is the 'missing link' that transforms a simple photo into a powerful tool for global health management.
 - The Global Health Burden
 - The "Accuracy Gap" in Human Perception
 - Data-Driven Nutrition

Background Study: Some Existing Applications and Limitations

- **MyFitnessPal, Lose It!** – **Manual** logging, high user effort
- **BiteSnap** – Single-image estimation, **weak portion accuracy**
- **Foodvisor** – Automated but **opaque** estimation
- **General Observation** – The most accurate features are often locked behind **paid** subscriptions

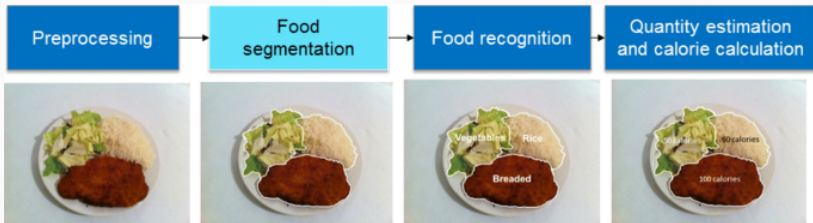
Background Study: Automated Food Calorie Estimation

- **Liang & Li, 2017** – Multi-view + Faster R-CNN, **volume error $\leq 30\%$**
- **Chang, 2013** – Multi-view, **volume error 5–10%**
- **Thames et al., 2021 (Nutrition5k)** – RGB+Depth regression, **AI often outperforms human experts**
- **Shao et al., 2023** – Single-view 3D shape reconstruction, **MAE 40 kcal, MAPE 11%**
- **Zhou et al., 2018** – Human estimation study, **accuracy $\geq 30\%$ within 20% of true calories**

Background Study: A Paper

- **Title:** Measuring Calorie and Nutrition From Food Image
- **Authors:** Parisa Pouladzadeh, Shervin Shirmohammadi, Rana Almaghrabi
- **Published in:** IEEE Transactions on Instrumentation and Measurement, Vol. 63, No. 8, Aug 2014
- **Data Used:** 3000 images (150 training + 150 test per class $\times$ 10 classes) — multi-condition (lighting, cameras, angles)
- **Recognition Accuracy:** Combined features **92%**
- **Calorie Estimation Accuracy:** Approx. **86% average** compared with real measured values
- **Limitations:** Lower performance on mixed food; segmentation errors; assumes constant depth for volume

Background Study: Method Overview Diagram



- Image segmentation extraction (color, texture, size, shape)
- Food recognition via SVM classifier
- Volume estimated from segmentation → mass via density tables → calories via nutritional tables

Background Study: Existing Methods

- **Manual Logging**
 - High accuracy, very high user effort
- **Single-Image (Monocular) Methods**
 - Low effort, poor volume estimation
- **Multi-View Image Methods**
 - Better geometric consistency
 - Balanced accuracy vs effort
- **Depth Sensor–Based Methods**
 - High accuracy
 - Poor scalability

- Image-based dietary assessment relies on:
 - Food detection and classification
 - Portion size estimation
- **Volume estimation** is more challenging than recognition
- Most systems rely on geometric assumptions or multiple views

Background Study: Existing Solution Categories Summary

- **Single-image (monocular):** Estimates food volume from one image using visual cues; simple but less accurate
- **Two-image methods (top + side view):** Two-image methods (top + side view): Combines top and side views to estimate height and area; more accurate than single image.
- **Multi-view methods (multiple side views):** Uses multiple angles to reconstruct 3D shape; higher accuracy with more computation.
- **Depth sensor-based approaches (LiDAR / RGB-D):** Uses depth sensors to directly capture 3D structure; most accurate but needs extra hardware.

Our Proposed Solution Overview

Proposed Solution: System Overview

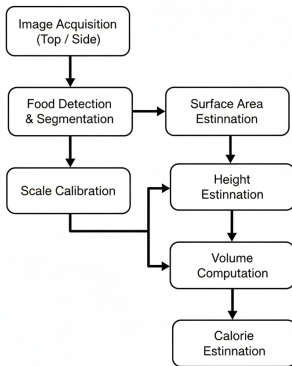
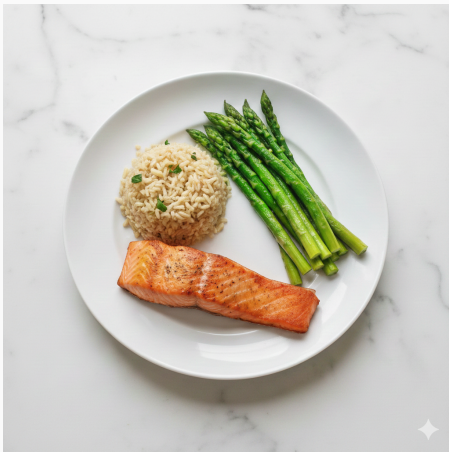


Figure 2: System overview

Step 1: Top View Image Acquisition

- User captures a top-down image of the meal
- Thumb or known object used for scale reference
- Provides clear visibility of food boundaries



Step 2: Food Detection and Segmentation

- YOLOv8-seg used for:
 - Food classification
 - Pixel-wise segmentation
- Segmentation masks enable precise area estimation



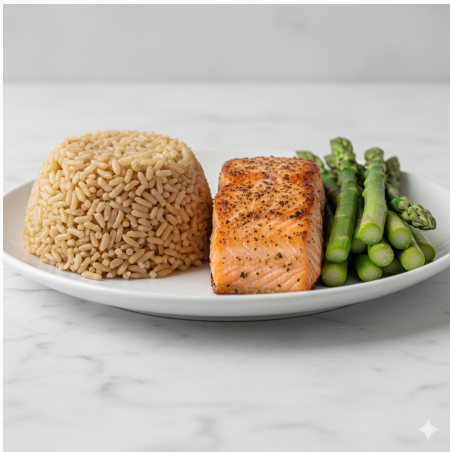
Step 3: Surface Area Estimation

- Segmented food regions converted to real-world scale
- Pixel-to-metric conversion using thumb reference
- Accurate estimation of food footprint area

$$\text{Surface Area} = N_{\text{pixels}} \times \text{scale factor}$$

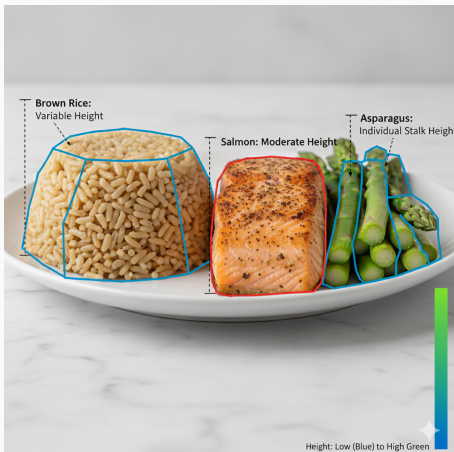
Step 4: Side View Image Acquisition

- Side view captures vertical structure of food
- Plate surface detected and used as baseline
- Resolves height ambiguity from top view



Step 5: Height Estimation from Side View

- Food silhouette extracted from side image
- Vertical distance measured from plate baseline
- Maximum visible height used as height estimate



Step 6: Volume and Calorie Estimation

- Volume approximated using geometric extrusion
- Food-specific density used for calorie estimation

$$\text{Volume} = \text{Surface Area} \times \text{Estimated Height}$$

$$\text{Calories} = \text{Volume} \times \text{Density} \times \text{Energy Factor}$$

Proposed Approaches

- **Approach A:** Single oblique image
- **Approach B:** Top view + one side view
- **Approach C:** Top view + multiple side views
- All approaches share the same pipeline but differ in:
 - Number of images
 - Robustness of height estimation

Proposed Approach A: Single 45° Image

Input: One oblique (45°) image with thumb calibration

- Single-view solution designed for minimal user effort
- Attempts to estimate both surface area and height from one image
- Most lightweight but least stable approach

Approach A: Single 45° Image

Input: One oblique (45°) image containing food and thumb

- Single-image solution designed for minimal user effort
- Both surface area and height inferred from the same view
- Fast but sensitive to perspective and occlusion

Approach A: Processing Pipeline

1. Image Acquisition

- User captures one oblique image with visible thumb

2. Food Detection and Segmentation

- YOLOv8-seg performs food classification and pixel-wise segmentation

3. Scale Calibration

- Thumb detected in the image
- Known thumb width used to compute pixel-to-metric scale

4. Surface Area Estimation

- Table plane detected using Canny edge detection and Hough line transform
- Perspective homography applied to rectify the table surface
- Food mask projected onto rectified plane to compute real-world area

5. Height Estimation

- Visible vertical extent of the food used as height proxy
- Assumes upright food and minimal self-occlusion

Approach A: Volume Estimation and Limitations

$$\text{Volume} = \text{Area} \times \text{Height}$$

- Area and height are both derived from a single viewpoint
- Errors in scale or perspective propagate multiplicatively
- Highly sensitive to camera angle, viewpoint, and occlusion

Approach B: Top View + One Side View

Input: Two images (top view and side view), thumb visible in both

- Separates surface area and height estimation
- Improves robustness over single-view approach
- Moderate increase in user effort with significant accuracy gain

Approach B: Processing Pipeline

1. Image Acquisition

- User captures one top view and one side view of the meal
- Thumb included in both images for scale calibration

2. Food Detection and Segmentation (Top View)

- YOLOv8-seg applied to top view image
- Produces accurate pixel-wise food masks

3. Scale Calibration

- Thumb detected in top and side views
- Pixel-to-metric scale computed independently for each view

4. Surface Area Estimation

- Food mask from top view converted to real-world area
- Top-down geometry minimizes perspective distortion

5. Height Estimation (Side View)

- Plate detected as horizontal baseline
- Food silhouette extracted above the plate
- Maximum vertical extent used as food height

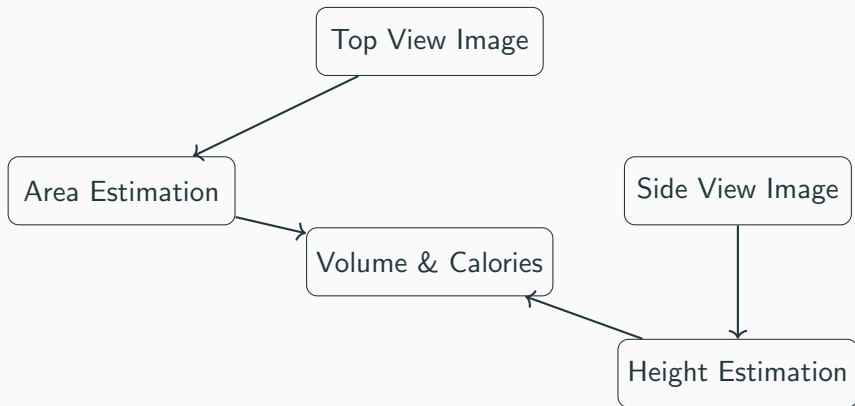
6. Volume and Calorie Computation

Approach B: Volume Estimation and Characteristics

$$\text{Volume} = \text{Area}_{\text{top}} \times \text{Height}_{\text{side}}$$

- Area and height are estimated independently
- Significantly more stable than single-image estimation
- Errors are less correlated across views

Approach B: Conceptual Diagram



Approach C: Top View + Four Side Views

Input: One top view and four side views captured at different angles

- Multi-view solution designed for maximum robustness
- Handles irregular food shapes and self-occlusion
- Highest accuracy at the cost of increased user effort

Approach C: Processing Pipeline

1. Image Acquisition

- User captures one top view with thumb
- Four side views captured approximately 90° apart

2. Food Detection and Segmentation (Top View)

- YOLOv8-seg applied to top view image
- Produces accurate food segmentation masks

3. Scale Calibration and Transfer

- Thumb used to establish scale in top view
- Plate geometry extracted from top view
- Scale transferred to side views using known plate dimensions

4. Surface Area Estimation

- Area computed from top-view segmentation
- Identical to Approach B for consistency

5. Multi-View Height Estimation

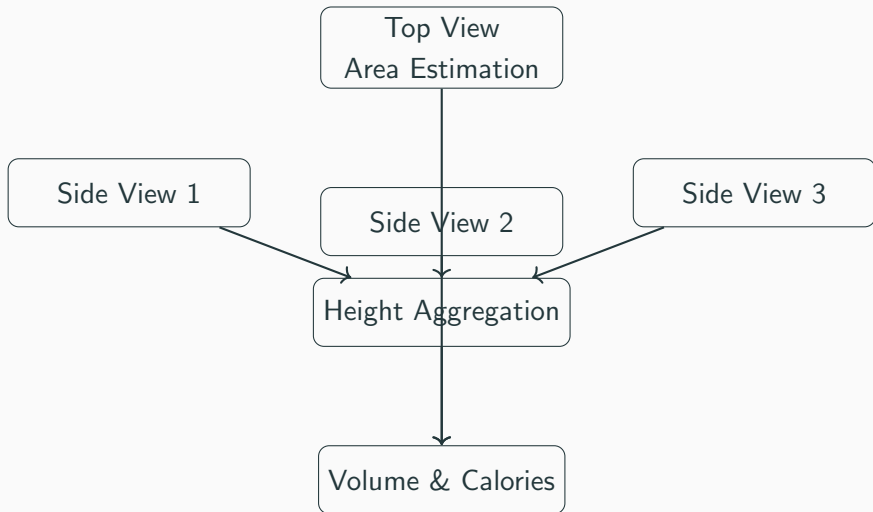
- Plate baseline detected in each side view
- Food silhouette extracted independently
- Height estimated separately from each viewpoint

Approach C: Volume Estimation and Characteristics

$$\text{Volume} = \text{Area}_{\text{top}} \times \text{Aggregated Height}$$

- Most robust volume estimation among all approaches
- Reduced sensitivity to occlusion and viewpoint bias
- Trades user effort for improved reliability

Approach C: Conceptual Diagram



Comparisn between approaches

Comparison of Proposed Approaches

Table 1: Comparison of Food Volume Estimation Approaches

Approach	Accuracy	User Effort	Advantages	Disadvantages
A: Single-Image (Monocular)	Low	Very Low	High User Adoption: Fast and seamless; works with any standard smartphone camera without special setup.	Depth Ambiguity: Highly prone to errors as the system must "guess" the Z-axis (depth) without a reference.
B: Two-Views (Top + Side)	Medium	Medium	Better Spatial Context: Two angles provide a clearer definition of food boundaries and reduce occlusion.	User Friction: Requires the user to take multiple specific photos, which can be inconvenient in social or public settings.
C: Multi-View	High	High	Precise Reconstruction: Captures complex, irregular shapes accurately for clinical-grade volume estimation.	Impracticality: Extremely tedious for daily meal logging; requires significant time and high computational power.

Approach B: Key Advantages

- Decouples area and height estimation
- Lower error amplification
- Balanced accuracy and user effort
- Strong geometric justification

Approach C: Key Advantages

- Most robust height estimation
- Handles occlusion and irregular shapes
- Lower variance and tighter error bounds
- Highest expected accuracy

Economic Impact Analysis

- **No specialized hardware required:** Uses existing smartphones, reducing upfront costs and enabling low-cost deployment.
- **Reduction in Direct Healthcare Expenditures:** Supports better dietary monitoring, helping prevent diet-related diseases and lowering treatment costs.
- **Indirect Economic and Productivity Gains:** Improved health outcomes reduce absenteeism and increase overall workforce productivity.
- **Scalability:** Easily scalable to large populations with minimal additional cost, increasing long-term economic impact

Environmental Impact Analysis

- **Reduction in Food Waste:** Accurate portion and calorie estimation helps users serve and consume appropriate amounts, reducing food waste.
- **Lower Resource Consumption:** Uses digital, image-based methods instead of physical measuring tools, reducing material and resource usage.
- **Reduced Carbon Footprint:** Minimizes the need for additional hardware manufacturing and transportation by relying on existing smartphones.
- **Support for Sustainable Dietary Habits:** Encourages mindful eating and healthier food choices, contributing to more sustainable food consumption patterns.
- **Scalable Eco-Friendly Solution:** Software-based deployment allows large-scale adoption without significant environmental overhead.

Social and Ethical Considerations

- **User Privacy and Data Protection:** Food images may contain personal or contextual information, requiring secure data handling and privacy-preserving methods.
- **Informed User Consent:** Users should be clearly informed about how images and data are collected, stored, and used.
- **Bias and Fairness:** Models trained on limited datasets may perform poorly across different cuisines, cultures, or portion styles.
- **Accessibility and Digital Inclusion:** Smartphone-based solutions improve accessibility but may still exclude users with limited digital literacy or access.
- **Psychological and Social Impact:** Over-reliance on calorie tracking may affect eating behavior, requiring careful and responsible system design.